

17 — Infeasible Plant: ME Too Small (expected 400, no results)

Proves · Validation refusing the plant before Level 1 ever runs — a different failure path from 09.

Mechanics this scenario turns on

- Validation runs **before** Level 1. A plant that cannot carry its load is rejected there, with a different message and a different code path from the Level 1 rejections.

Panels described below · The rejection arithmetic · Two different guards — compare with test 09

Anything not described here behaves exactly as in `01-excel-baseline` — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 09.

Excel loads (11 463 / 3 800) but ME **2×5 000**, SG 0, AE 4×4 000, no battery. Supposed to fail — the red banner is the pass criterion.

The rejection arithmetic

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Installed ME = 2 × 5 000 = 10 000 kW < propulsion demand 11 463 kW
→ ME utilization would exceed 100 % no matter which combination runs
→ INPUT VALIDATION promotes the capacity warning to an error (before Level 1 even starts)
→ HTTP 400: "Main engine utilization > 100%. Consider reducing propulsion power, decreasing
    sea margin, reduce hotel/mission load or increasing main engine capacity."
```

Two different guards — compare with test 09

	Test 09 (PTI 50)	Test 17 (ME 2×5 000)
Where it fails	Level-1 enumeration (battery-PTI gate kills every combo)	Input validation, before L1
Message	battery needs 200.6 kW PTI, only 100 available...	ME utilization > 100 %...
Physics	plant can carry the load, battery has no path to the shaft	plant simply too small

Both: HTTP 400 (user problem, not server fault), empty results panel, actionable message listing the levers that would fix it (note "decreasing sea margin" — the SM chain from test 16).

Takeaway: the app distinguishes WHY a configuration is impossible and says so with the actual numbers — no generic "error occurred".